

Haashir Mohammed

Mrs. Ireland

Independent Research G/T

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Man-made measures:

Using AI to Augment Gene And Stem Cell Therapies to treat Parkinson's Disease

Abstract

Parkinson's Disease (PD) is a neurodegenerative disease which currently cannot be treated with a cure. Both gene and stem cell therapies propose new solutions but have to this point failed to arrive at the same stage as a standard of care; however, the use of AI has advanced the precision of these potential treatments. The work of Mathis et al. (1151), Witten et al. (1790), Singh et al., and Garmany and Terzic (613) demonstrate the AI model's ability to make accurate predictions. However, although these are exciting advances it would be too soon to integrate AI in a clinical setting without further work, because: There are no large, comprehensive, standardized data-sets, data-sets already available contain bias, there can be a "black-box effect" (where the reasoning behind AI's outcome cannot be easily justified or understood), and the absence of human clinical trials. Overall, there is real potential in using AI to advance gene and stem cell therapies but no timeframe can yet be given and all the gaps detailed in this review need to be solved before AI-augmented treatments are adopted in a clinical environment.

Introduction

According to the Parkinson's Foundation, there are 90,000 people who are diagnosed with Parkinson's Disease in the United States each year. It is the fastest growing neurological disease in the world, and the number of cases globally is expected to reach 12 million by 2040. Artificial Intelligence may be used to advance and hasten research to diagnose and cure Parkinson's, as it can analyze and summarize large swathes of data, and mimic human intelligence with regard to learning and problem-solving. However, there are problems that hinder development and implementation of AI in the medical workspace to be used to effectively combat such conditions as Parkinson's, as it requires precision treatment, and AI is currently not proficient enough to fulfill this criterion. Nevertheless, AI's aptitude as an intelligent machine cannot be denied. Although there are many social, ethical, and legal obstacles that need to be overcome, as well as the fact that AI is not fully sophisticated and refined, due to its now capable data analysis abilities, it has the potential to be more effective paired with human experts in chronic disease research, particularly neurodegenerative diseases such as Parkinson's. The purpose of this paper is to review some of the most recent developments concerning AI treatments for PD based on five key criteria: (1) the ability of AI to predict the outcomes of gene therapy procedures based on training data, (2) the ability of AI to predict the outcomes of stem cell therapy culture development based on training data, (3) the environment of the AI's training and study and whether the model's accuracy could translate to a clinical setup, (4) the ethical and social implications associated with using AI to develop treatment for disease, and (5) other drawbacks regarding AI such as a lack of large, comprehensive datasets and the "black-box problem."

Review of Literature

Parkinson's is a neurodegenerative disease caused by the deterioration of dopaminergic neurons in a patient's brain, manifesting symptoms such as bradykinesia, tremors, rigidity etc. AI, or Artificial Intelligence, are machines that mimic human intelligence, ability to learn, problem-solving, pattern recognition, and decision-making. Since its discovery in 1817, Parkinson's Disease, or PD, has not had a cure. Experimental treatments exist, however none are effective consistently. AI can access the entire internet to solve problems faster, making it able to elevate and augment clinicians care (Jackson). It can also analyze very large datasets quickly and help clinicians make more timely decisions on how to carry out diagnosis or treatment, saving time that would otherwise be spent on research (Ellis). AI has demonstrated accuracy in diagnosing and detecting PD, able to detect abnormalities in gait, handwriting or facial expression (Wu et al.; Yang et al.). AI has also shown proficiency in data analysis and predictions in the fields of possible treatments for PD. The two possible treatments for Parkinson's Disease with the most potential to be effective are Stem cell therapy and Gene therapy; however, neither of these experimental treatments are qualified to be implemented in the medical workspace.

Stem cells are cells that can differentiate into different types of cells and take on their function. Using stem cells in a PD patient's brain can regenerate the damaged dopaminergic neurons in it, effectively curing the disease. Using AI to analyze stem cells can help experts achieve a better understanding of stem cell behaviour, differentiation, classification, and modelling (Choudhery et al.). More research is needed for stem cells, as there are differences in potency and function, depending on sourcing, method of harvest, and delivery; there are also obstacles in scaling and standardization (Mukherjee et al.). It is necessary for stem cells to be

grown in large quantities for treatments to be viable; however, producing these stem cells at a large scale is inefficient and expensive. There is a lack of standardization in the methods of growing, handling, and preparing stem cells for treatment. Different studies will use different methods, leading to varied results (Mukherjee et al.). Manufacturing stem cells also entails high labor costs. There is difficulty in automating the process of manufacturing and producing stem cells because each group may contain differences in growth rate, shape, color or colony density, which may require extemporaneous actions. Overall, while stem cells demonstrate potential in treating PD, and AI demonstrates proficiency in augmenting stem cell therapy's development, neither are advanced enough to elicit the creation of a timeline of implementation into the clinical environment.

According to ("What Role Does AI Play"), "gene therapy is a therapeutic strategy that involves the deliberate modification of an organism's genetic material to prevent or treat disease." There are multiple types of gene therapy, considering what action is performed on the gene, such as gene complementation, silencing, supplementation, and editing, wherein an existing gene is supplemented with a new one, a harmful gene is removed, a gene is added, or a mutation in an existing gene is permanently repaired (Szunyogh et al.). In order to carry out gene therapy, a "delivery vehicle" called a vector is required. Viruses are the most commonly used vectors, as they are naturally effective at entering cells. The most commonly used virus types are Adeno-Associated Virus (AAV), Lentivirus, Adenovirus, and Herpes Simplex Virus (HSV) (Szunyogh et al.). Multiple different types of vectors exist, however, each has limitations. For example, AAVs can only carry small genes efficiently and are refined in their production, whereas HSVs can carry large genes, but are less developed clinically. There is also a lack of consistency regarding delivery itself. While most trials do involve injecting the virus directly, the

rate, volume, and exact placement of it varies between studies, and intravenous placement is still being explored (Szunyogh et al.). There are also inconsistencies with precision targeting (Tan et al.). AI has provided a new frontier in AAV vector research, and gene therapy as a whole. AI's pattern recognition talents may allow it to predict whether a vector may be an effective delivery agent (Tan et al.). While this AAV engineering research is still in its early stage, it provides a new world of potential in gene therapy; using AI has already been proven to augment gene therapy and has shown remarkable progress in editing and rewriting the gene (Tan et al.; Dixit et al.).

Despite the progress that AI has made and the proficiency it has in predicting the outcomes of gene therapy and stem cell therapy, there are multiple issues to address in both areas of treatment. On the forefront of these issues is the lack of large, comprehensive datasets that AI requires to prevent bias. In order to possess diversity, it is imperative that AI be trained on data from an entire population, a large range of people to give it unbiased training (Matheny et al.). Being too general can lead to failed learning across demographics, creating inequalities due to biased training (Minesh 142). Another obstacle is data collection. Data collection is crucial, and AI requires training using supervised learning, and then optimization using a human to make the model streamlined and sophisticated, making it capable of completing complex tasks (Dixit et al.; Saxena). The size of the dataset is also crucial, as small datasets can lead to inept models that are not trained properly, which can lead to inaccurate diagnosis and treatment (Yang et al.). Models can also “overfit,” that is, they are able to memorize and use the data they are trained on, but fail on any new data that is provided (Tan et al.).

While data collection and datasets are the forefront of the hurdles regarding AI, another prominent issue plaguing the AI community is the “black-box” problem. The black-box problem

refers to the inability of humans to understand AI's decision making process and reasons for producing a certain output. This creates a lack of trust in professionals. The main solution to this problem is to have a human in the loop, monitoring the AI and making sure it does not hallucinate, because AI is not capable of advanced processes yet. Using "red-teaming" is recommended, a process in which, over a period of a few months, the AI is given "doomsday" scenarios to see how it responds, and whether that response is what the team that trains the AI desires and expects (Saxena). Using guardrails are the primary method of regulating AI, and using the different types of AI to regulate: the content AI can generate, how the AI behaves and responds, what information the AI possesses, what it can expose and reveal, its authorization for access to sensitive information, and what content AI can generate in regards to business or industry rules and needs. These are the most feasible ways to control AI, and to ensure humans can control AI without it becoming unmanageable (Saxena).

Along with the obstacles pertaining to data and hallucinations, there are also ethical concerns regarding using AI. The foremost is data privacy; due to the need for data security and privacy protection, AI requires guardrails to be prevented from releasing this information to unauthorized individuals and also prevent it from accessing the internet (Saxena). This can be done by using guardrails to curb AI's creativity and power, redacting information the model does not require. While this decreases AI's effectiveness, it is a necessary measure that must be implemented to prevent the aforementioned undesirable actions. The model needs to be treated like any other human employee, using regular audits and checks (Saxena). Humans must also play a role by participating in data "stewardship," using secure storage and encrypted transmissions, restricting access to information; participants and patients should also be informed of what using AI entails and give educated consent (Martinez-Martin et al.). Data diversity also

plays a role, as bias in the AI model is an ethical concern as well as a quality concern. AI needs to be trained through supervised and third-party learning to have minimal bias (Martinez-Martin et al.). These measures and precautions are necessary because it is known and proven to perform unsolicited and uninformed diagnosis (Youngmann et al. 2503). Due to such a possibility, responsibilities extend to technology companies from clients when implementing AI in the workplace (Youngmann et al. 2503).

Prior to 2022, AI was not considered advanced enough to complete complex tasks in a wide array of fields. However, following the AI “boom” and advent of Large Language Models (LLMs), it now possesses remarkable generative capabilities and multimodal flexibility. Now, due to its now capable data analysis abilities, it has the potential to be effective paired with human experts in chronic disease research. In the next decade, a treatment for Parkinson’s can become effective and mainstream, paving the way for other conditions and diseases previously thought to be incurable, to also be treated.

Research Methods and Data Collection

The data collection process for this meta-analysis was designed to ensure a comprehensive and systematic gathering of relevant studies aligned with the research objectives. A structured search strategy was employed across multiple peer-reviewed databases to identify studies that met the inclusion criteria. Each selected study was carefully screened for methodological rigor, relevance to the research question, and the availability of qualitative data suitable for synthesis. Each study was analyzed for 4 themes: AI’s ability to bring about technological advancement, AI’s limitations in predicting power, ethical and social implications, as well as evidence for a lack of comprehensive datasets and the presence of the black-box problem. Four papers were used: The first discussed the efficiency of a deep learning algorithm

trained on prime editing data, the second discussed Artificial Intelligence-guided design of lipid nanoparticles for pulmonary gene therapy, the third discussed AI's predictive power regarding stem cell cultures, and the fourth discussed AI's applications in regenerative medicine. Each study used Neural Networks, and some included other types of Machine Learning and Deep Learning models such as GANs. Each study was conducted in an in silico, in vitro, and animal in vivo setting. While a majority of these studies are not pertinent or related to neurological conditions such as this paper's subject, it demonstrates AI's strengths and limitations, which can be re-applied to such disorders and diseases.

Results

Predicting prime editing efficiency and product purity by deep learning

The article focuses on using AI to augment the efficiency of prime editing. Prime editing is a refined version of the existing gene editing tool CRISPR-Cas9, which would “cut out” unwanted genes, but not repair them. Once a deletion of a gene was made, the correct and healthy repair of the gene could not be ensured. Prime editing refines this process by searching for the specific gene that needs to be deleted and “re-write” that gene exclusively, without damaging the rest of the cell. The conductors of the study used an RNN named PRIDICT. It displayed a high accuracy of 89% at correcting genes. No human trials were conducted, though they were planned to take place in the near future. While the study used a very large pegRNA dataset, in silico, in vitro, and mice in vivo data comprised the entirety of it. There was a lack of discussion on the social and ethical implications of AI, but a lack of data and the black-box problem was addressed, by analyzing the XGBoost model, a classic machine learning model, with SHAP analysis, and used integrated gradients to analyze PRIDICT.

Artificial Intelligence-Guided Design of Lipid Nanoparticles for Pulmonary Gene Therapy

Despite its name, gene therapy cannot be conducted using only genes. If genes were injected alone into the body, the immune system would destroy them before they could carry out their assignment. Thus, the conductors of the study used lipids as a delivery agent so that the gene could get to the cell safely. Scientists trained an AI to predict lipids that would be best suited as delivery agents. The AI's Pearson Correlation Accuracy (PCA) score was between 0.6 and 0.9. However, some datasets had a PCA score between 0.2 and 0.3. The scientists also only trained the AI on in vitro and ferret in vivo data, without human trials. There is no mention of ethics or a lack of comprehensive datasets. However, black-box problem is addressed.

AI-Driven Quality Monitoring and Control in Stem Cell Cultures: A Comprehensive Review

Analyzing stem cells is a long, inefficient, time-consuming process. Currently, scientists use Microscopy, Flow cytometry, and Immunostaining to analyze stem cells. However, these methods are either too slow or destroy the sample. This paper argues that using AI can augment the analysis of stem cells to streamline the process. A “unified framework linking stem cell culture attributes with AI-driven quality control” was used (See figure 1A). A collection of models, including CNNs, GANs, SVMs, and reinforcement learning, as well as sensor data, were used to analyze different aspects of stem cell culture manufacture (See figure 1B). The AI can identify healthy stem cell colonies, predict contamination, improve cell growth efficiency, and even produce its own training data (this is because there is usually a lack of comprehensive, large datasets) by using GANs to create more training images. Using reinforcement learning, cell growth efficiency improved by 15-18%. However, this success does not come without context and limitations. Every instance of high AI accuracy came from in vitro, in silico, and animal in vivo studies. The bottom line is that while AI has the potential, there is no prediction or likely

time that AI is confirmed to be good enough to implement in the healthcare space. There are also ethical and social implications such as privacy and consent. There is also a prevalence of data heterogeneity, as well as bias in these datasets due to an overrepresentation of certain demographics. Models can also overfit. The black-box problem is also prevalent. To address the black-box problem, the conductors of the study explored Explainable AI (XAI) systems to avoid it, noting that they had already been effective at chromatographic protein purification processes. For stem cells, the conductors pointed toward attention mechanisms and feature attribution as techniques that enable models to highlight which morphological or molecular features influence quality assessments, which facilitates human validation and eases regulatory review. XAI is illustrated as converting ML data into an explainable model with an explainable interface to be better understood by humans (See figure 1C).

Artificial Intelligence Powers Regenerative Medicine into Predictive Realm

This paper covers AI's capabilities in discovery of new regenerative treatments, manufacturing such treatments, and implementing them in a clinical setting. AI is successful at all of these. Again, the AI's predictions were only based on lab and animal data and no human trials. The paper admits AI in regenerative medicine is still new and unexplored. AI is promising but unproven. Ethical and social implications include equitable access across financial categories, data privacy, and governance (the chance of a monopoly coming in with only one group controlling the regenerative AI market). There is also a lack of comprehensive, large datasets, bias in datasets, and the black-box problem.

Conclusions

Parkinson's disease is one of the most significant issues of 21st-century medicine and both gene therapy and stem cell therapy have emerged as two of the most promising

experimental therapies, each research being conducted into a potential cure. As yet, neither approach has developed into the consistent and reliable methodology required for clinical application. However, artificial intelligence has played the role of an indispensable supportive technology. As well as accurately predicting the efficiency of prime editing and optimizing delivery vectors it can monitor the stem cell cultures, and its role is crucial in accelerating the development process of regenerative therapies. The four studies which make up this review have revealed that the prediction of AI is already a quantifiable, measurable phenomenon, with accuracy in gene editing prediction and cell growth optimization in stem cell cultures reaching 89% and 15–18%, respectively.

However, these findings must be accompanied with a serious dose of skepticism; in every case of AI accuracy in the studies mentioned above they were achieved under *in silico*, *in vitro*, or animal *in vivo* environments and not through a single human clinical trial that would enable the validation of the results in the living body of Parkinsonian patient. Lack of standardized and demographically large and representative dataset prevent models to learn biases and accurately reflect patient heterogeneity and black-box problem, the lack of transparency in the algorithms, still remains as a serious obstacle to its trust. The issue of ethical concerns, in particular data privacy, equity and governance have not been addressed by any studies in this review.

Future directions for this research include moving on to human clinical trials, to further refine AI's processing and training, by using explainable models despite reduced performance, and large, enhanced datasets.

What is concluded from this review is not an inefficient performance of AI, but an immature science in its initial and important stage; a field which requires further infrastructure, standard and representative datasets, effective regulatory mechanisms and robust mechanisms of

human supervision that must be developed together with technology and not after. Human clinical trials must take place, standards for datasets must be established, and the rationale behind the prediction of AI must be auditable before gene and stem cell therapies under support of AI can be used in clinical practice. Its capability, unlike its timeframe, is not under question.

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Appendix

Figures

AI-Driven Quality Monitoring and Control in Stem Cell Cultures: A Comprehensive Review (Singh et al.)

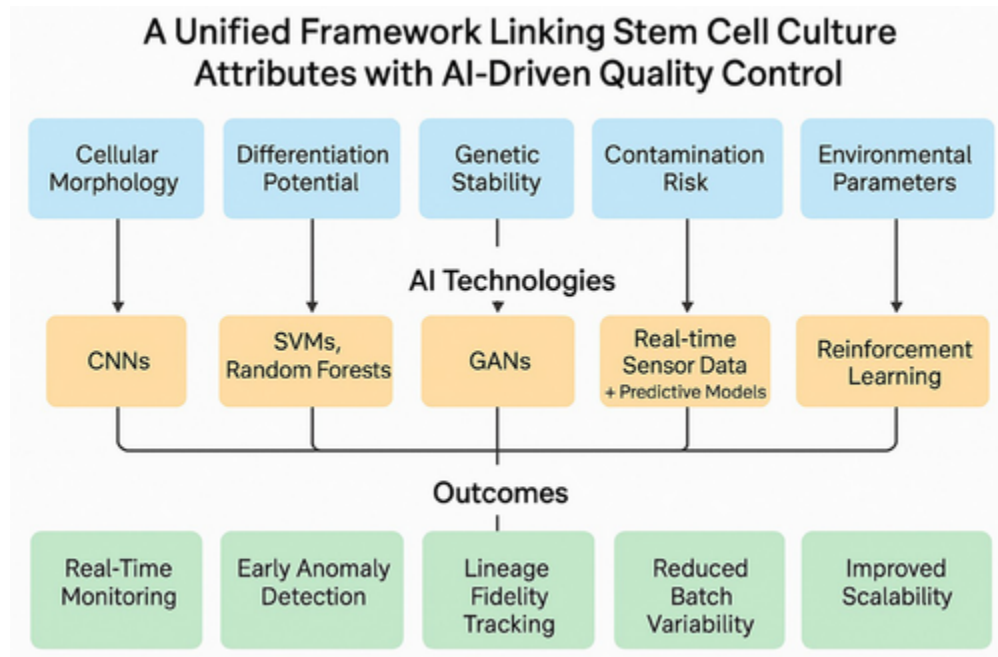


Figure 1A: Conceptual framework for AI-driven quality monitoring in stem cell cultures, linking critical quality attributes (CQAs) with corresponding AI technologies and control strategies. This framework structures the discussion in the subsequent sections (created using biorender.com)

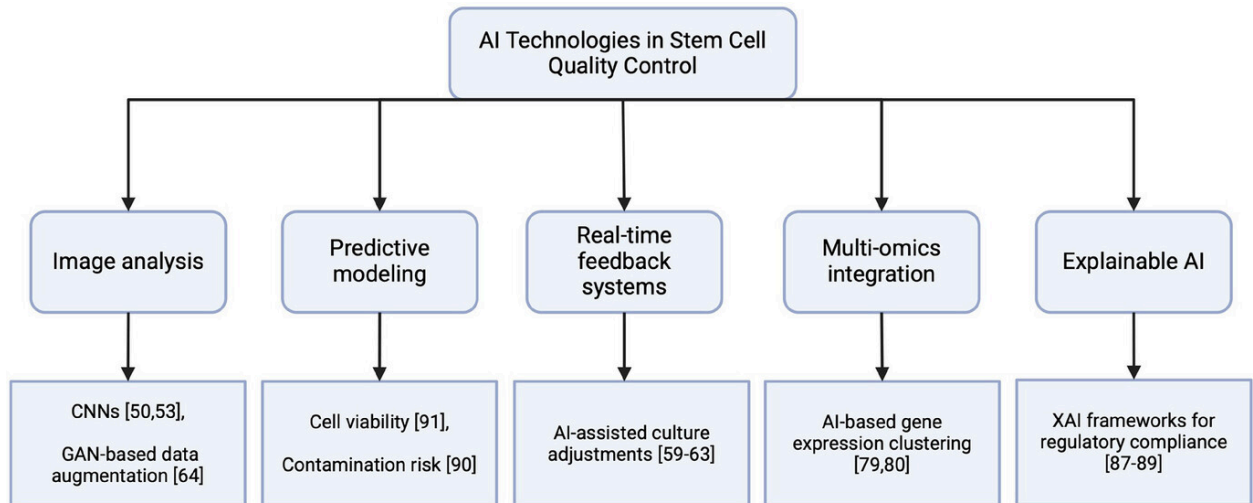


Figure 1B

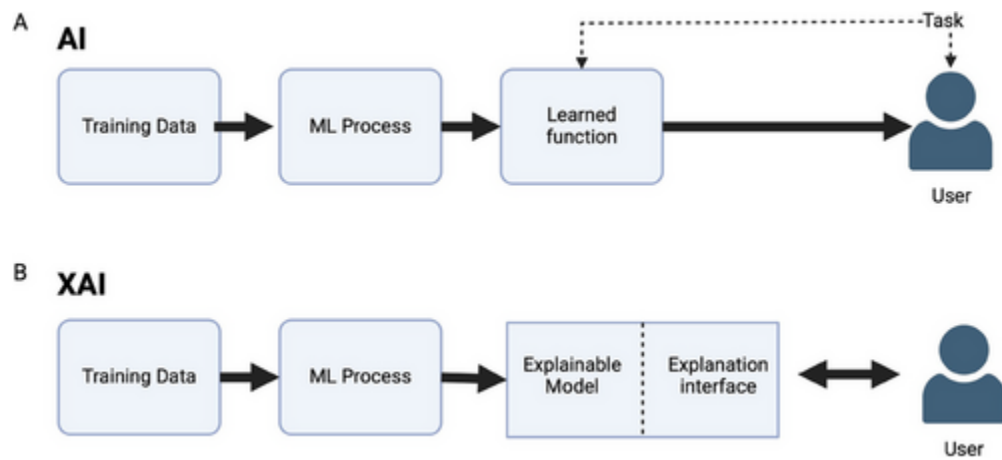


Figure 1C